Evidence Based Guidelines

Evaluation and Treatment of Kawasaki Disease

Guideline Title: Evaluation and Treatment of Kawasaki Disease

Scope: This document provides guidance on the acute diagnosis and management of Kawasaki disease (KD), based on the latest published data in current bibliography, but clinical decision making should be individualized as per specific patient circumstances.

Introduction:

- Kawasaki disease is an acute self-limiting febrile illness of unknown cause primarily affecting children < 5 years old.
- It is characterized by systemic inflammation in medium-sized arteries and multiple organs and tissues during the acute febrile phase.
- The aetiology of KD remains unknown, but most likely it represents an aberrant inflammatory host response to one or more as yet unidentified immunological triggers in genetically predisposed individuals.
- The most common cause of acquired cardiac disease in childhood.
  - 25% of untreated patients and 5% of treated patients developing coronary artery aneurysms.
  - Infants <6 months of age, the risk of a coronary artery aneurysm is 50%, even in KD patients who received treatment with IVIG within the first 10 days of illness.
- KD remains an important cause of long-term cardiac disease into adulthood.

Diagnosis of KD:

KD must be considered in any child with a febrile exanthemataous illness and evidence of inflammation, particularly if it persists longer than 4 days.

Complete Kawasaki (KD) (AHA 2017):

Fever of 5 days or more (the day of fever onset is taken to be the first day of fever), plus four of five of the following:

1. Conjunctivitis: Bilateral, bulbar, conjunctival injection without exudate
2. Lymphadenopathy: Cervical, often >1.5 cm usually unilateral
3. Rash: Maculopapular, diffuse erythroderma or erythema multiform
4. Changes of lips or oral mucosa: Red cracked lips, strawberry tongue or diffuse erythema of oropharynx
5. Changes to extremities: Erythema and oedema of palms and soles in acute phase and periungal desquamation in subacute phase

Diagnosing KD before day 5 of fever:

Diagnosis and treatment of KD should not be delayed if:

- Five out of six diagnostic criteria of KD are present before day 5 of fever
- Or if CAA (Z-score ≥2.5) or coronary dilatation (Z score >2, but <2.5) is present.

In the presence of ≥4 principal clinical criteria, particularly when redness and swelling of the hands and feet are present, the diagnosis may be made with only 4 days of fever.

Incomplete Kawasaki disease (Figure 1).

- Children with fever ≥5 days and two or three clinical criteria
- OR infants (<12 mo) with fever for ≥7 days without other explanation
- If coronary artery abnormalities are detected, the diagnosis of KD is considered confirmed in most cases.

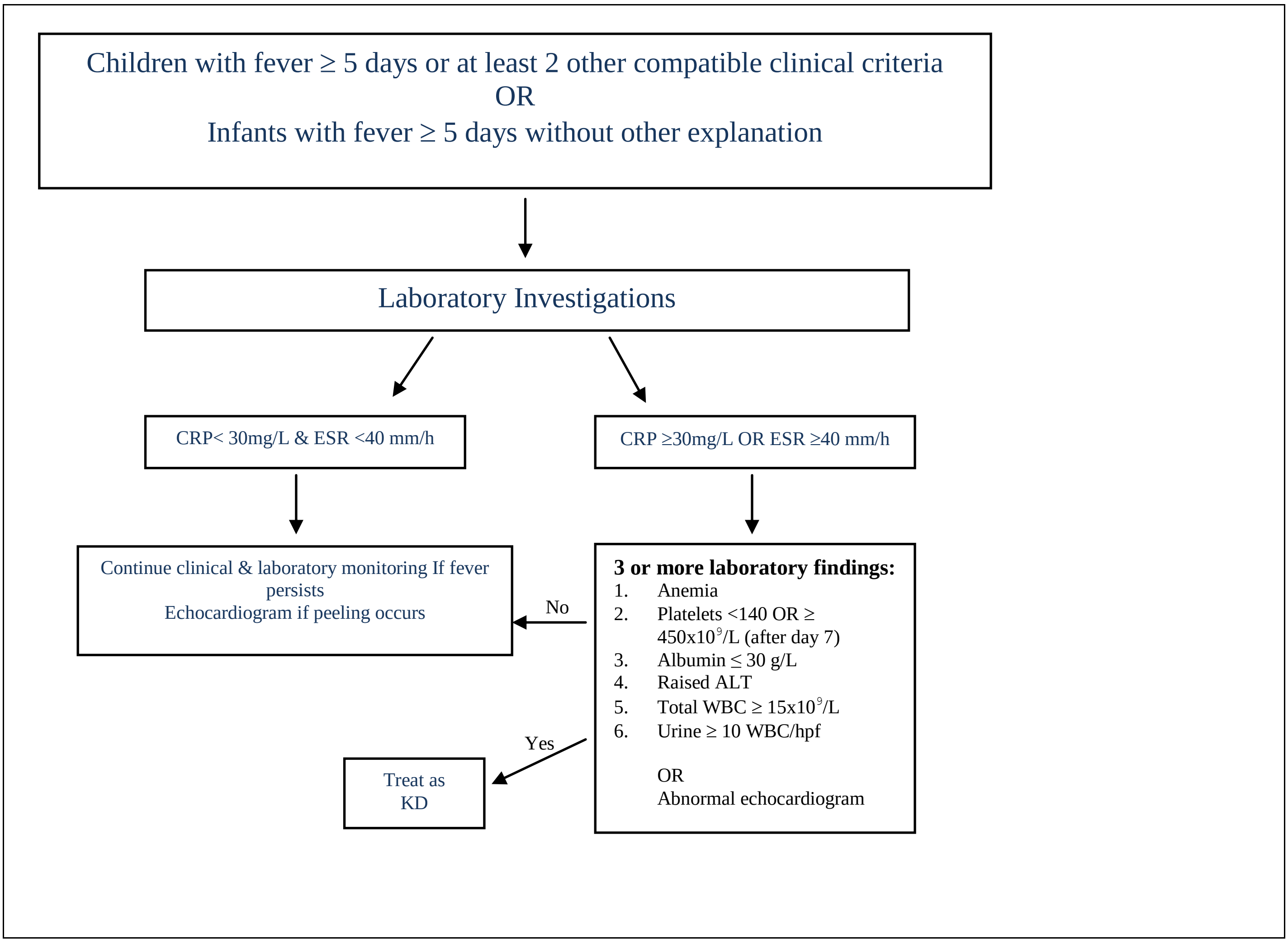


Figure 1. Evaluation of suspected Incomplete KD

Two clinical signs not included in the criteria are relatively specific to KD:

1. Irritability (maybe due to aseptic meningitis)
2. Erythema and induration at sites of BCG immunizations

Other Common Findings:

Arthritis, Aseptic meningitis, Pneumonia, Uveitis, Gastroenteritis, Urethritis/meatitis, hydrocele, Dysuria, Otitis, gallbladder hydrops, and Pancreatitis.

Signs and Symptoms Suggesting Disease Other than KD include:

Exudative conjunctivitis, Exudative pharyngitis, Discrete intraoral lesions or Bullous or vesicular rash or generalized lymphadenopathy.

Supplementary Laboratory Criteria Include:

Albumin < 3.0gm/dl (normal 3.4-5.4gm/dl), Anemia for age, Elevation of Alanine amino transferase, Platelets after 7 days >450,000/mm3 (normal 150,000-450,000/mm3), white blood cell count >15000/mm3 (normal 4,500 - 11,000/mm3) and Urine >10 WBCs/high power field (normal 2-5WBCs/hpf or less).

Other Differential Diagnoses to be considered:

- Toxic shock syndrome (Streptococcal and Staphylococcal)
- Staphylococcal scalded syndrome
- Scarlet fever
- Infection with Enterovirus, Adenovirus, Measles, Parvovirus, EB virus, CMV, Mycoplasma, Ricketssiae and Leptosprosis
- The DD of IVIG resistant KD includes Polyarteritis nodosa/ other vasculitidis, Systemic onset juvenile idiopathic arthritis and Malignancy (particularly Lymphoma).

Note:
A child with KD may have concurrent infection with a respiratory viral pathogen.
In a child with clinical findings compatible with classic KD, the detection of respiratory viruses such as respiratory syncytial virus, metapneumovirus, coronaviruses, parainfluenza viruses, or influenza viruses **does not exclude the diagnosis of KD.**

Management:

Treatment: Aimed at reducing inflammation and preventing the occurrence of CAA and Arterial thrombosis. Therapeutic goal is to **aim for "Zero fever, Zero CRP" as quickly as possible** (This means that 48 hours following complete IVIG treatment, patients should be afebrile and CRP should be either normal (<10mg/L) or at least halving every 24hours), so don't rely on resolution of fever alone as sole determinant of therapeutic success.

Initial Treatment:

- Acute treatment As soon as a patient is diagnosed with KD, treatment should be initiated. This applies to both complete and incomplete KD. **Do not delay treatment waiting for echo**.

- Initial treatment for KD and incomplete KD include IVIG and Aspirin. Approximately 80% of patients with KD respond to initial therapy.

- High risk cases should receive IVIG, aspirin, and corticosteroid. Those with the severe inflammation (e.g. CRP>100 mg/L) and the very young (<12 mo) are particularly high risk.
- For any patient considered to have features of high-risk disease, adding an adjunctive agent (glucocorticoids or non-glucocorticoid immunomodulatory agents) to be considered.

Immunoglobulins

IVIG 2 g/kg as a single infusion, usually given over 12 hours.

However, infants who have cardiac compromise may not be able to tolerate the fluid challenge associated with high dose and consideration of divided doses given over several days may be appropriate.

Note:

- ESR should NOT be used to assess response as it is accelerated by IVIG therapy
- A persistently high ESR alone should not be interpreted as a sign of IVIG resistance
- Adverse Effect: IVIG reaction, aseptic meningitis, fever, anaphylaxis.

Aspirin:

- Aspirin (ASA) antiplatelet dose of 3-5 mg/kg once daily. Continuation beyond 6 weeks is dependent on coronary status (guided by cardiology team)

Corticosteroids:

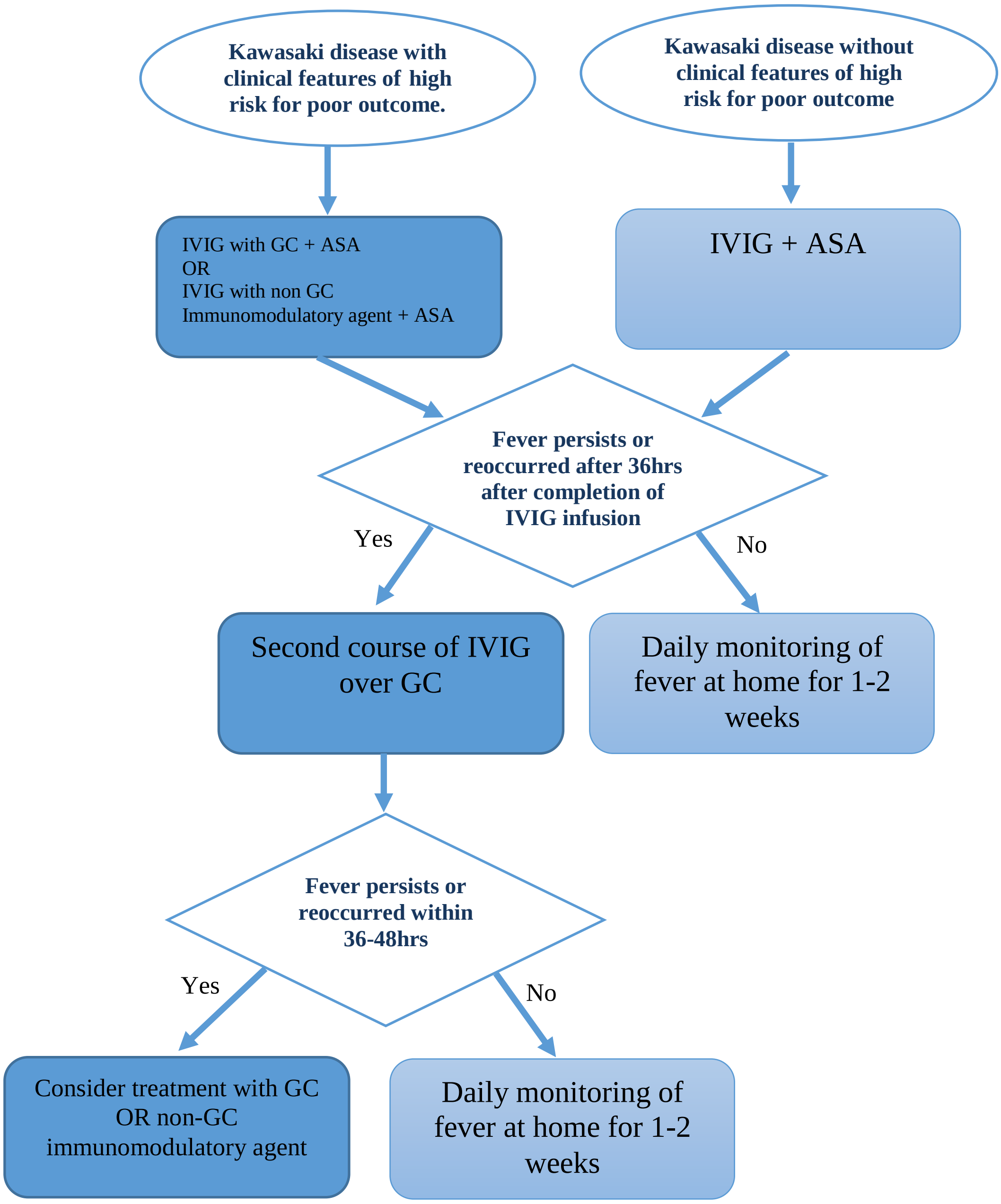
Should be given to patients with severe KD:

1. IVIG failure: ongoing fever, and/or persistent inflammation or clinical signs 48 hours after completion of first IVIG dose.
2. Severe disease: persistently elevated CRP, liver dysfunction, hypoalbuminaemia, anaemia, haemophagocytic lymphohistiocytosis (MAS), or shock.
3. Patients who already have evolving coronary and/or peripheral aneurysms with ongoing inflammation at presentation.

Suggested corticosteroid regimens:

- Methylprednisolone 0.8 mg/kg BD I.V. for 5-7 days or until CRP normalizes; then converts to oral prednisone/prednisolone 2 mg/kg/day and wean off over next 2-3 weeks.
- Methylprednisolone 10-30 mg/kg (up to maximum of 1g/day) once daily for 3 days followed by oral prednisone/prednisolone 2 mg/kg per day until day 7 or until CRP normalizes; then wean over next 2-3 weeks.

Figure 2: Algorithm for KD treatment



TNF-a blockade:

Should be considered in KD patients with persistent inflammation despite IVIG, aspirin and corticosteroid treatment, after discussion with rheumatology consultant. Infliximab 6/mg/kg is preferred anti-TNF agent for KD.

Anti-aggregation and Anticoagulation:

- **Medium sized aneurysms (Z score 5-10):** aspirin at 3-5 mg/kg/day and consider addition of an antagonist of adenine di-phosphate-mediated activation platelet aggregation such as clopidogrel, at a dose of 0.5-1.0 mg/kg/daily (Discuss with the cardiology team).
- **Giant aneurysms (internal diameter ≥8 mm, or Z-score ≥10, and/or coronary artery stenosis):** warfarin (in addition to aspirin), after initial heparinization; heparin can be stopped when a stable INR between 2-3 is reached. Low molecular weight heparin is a suitable alternative, particularly in young infants where safe warfarinization can be challenging

Specific Considerations:

- IVIG is ideally given in the first ten days of the fever
- It is reasonable to give treatment after 10 days if fever persists with ongoing inflammation (elevated CRP or ESR)
- Live vaccine should be avoided for 11 months after given IVIG
- NSAIDs should be avoided if possible because they interfere with the anti-platelet effect of aspirin and possible Reye's syndrome

Refractory Kawasaki disease (Figure 3):

There is no standard definition of refractory KD, however, most common definition is failure of fever to resolve within 36-48 hours after completion of first IVIG dose. A broader definition would be failure of resolution of systemic inflammation (fever, rapid fall in CRP) within 48 hours of primary treatment (IVIG/ASA +/- steroid).

Refractory KD Treatment Options: (Discuss with rheumatology team)

Start by giving IVIG second dose (2g/kg/day).
*Steroids if high risk patient < 12 months
If failed to respond then the following options are considered:

1. Corticosteroid Pulse (10-30mg/kg/day for 3 days) with tapering over 2 to 3 weeks as mentioned above.
2. Anti-TNF (Infliximab) single infusion 6-10mg/kg IV
3. IL-1 blockade (Anakinra) 2-8mg/kg/day until resolution of symptoms and improvement of inflammatory parameters.

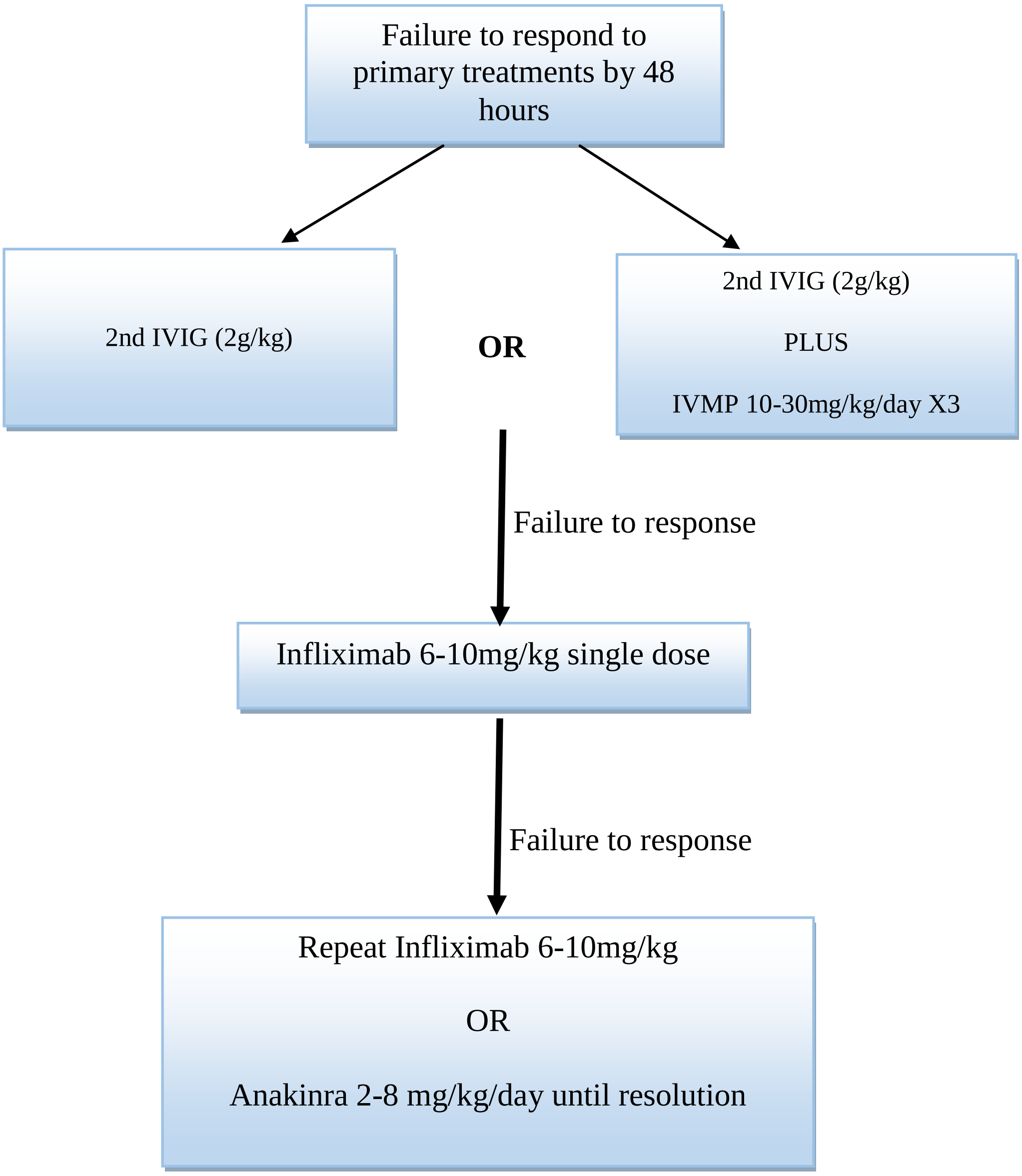


Figure 3. Refractory Kawasaki disease
*IVIG: immunoglobulins, IVMP: Intravenous Methylprednisolone

Links:

Useful related web sites

- **KD Foundation: http://www.kdfoundation.org/**
- **CDC: https://www.cdc.gov/kawasaki/**
- **healthychildren.org: - -**

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https://www.healthychildren.org/English/healthissues/conditions/heart/pages/KawasakiDisease.aspx

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